

Yann LeCun: Meta AI, Open Source, Limits of LLMs, AGI & the Future of AI | Lex Fridman Podcast [#416](#)

Yann LeCun, the chief AI scientist at Meta and a professor at NYU, is a proponent of open-sourcing AI development. He and Meta AI have open-sourced many of their models, including LLaMA 2, and plan to release LLaMA 3. LeCun criticizes those who warn about AGI as an existential threat, believing AGI will be benevolent and remain under human control.

LeCun has been involved in intense discussions about the future of AI, particularly around the role of autoregressive language models (LLMs) like GPT-4 and LLaMA. He argues that LLMs are not the path to superhuman intelligence because they lack essential characteristics of intelligent behavior, such as understanding the physical world, persistent memory, reasoning, and planning.

LLMs are trained on vast amounts of text from the internet, which seems like a significant knowledge base. However, LeCun points out that the sensory input a four-year-old child receives is far greater than the data LLMs are trained on. This suggests that most human knowledge comes from observation and interaction with the real world, not language.

LeCun debates the idea that intelligence must be grounded in reality, advocating for AI that can interact with and learn from a rich environment. He believes language is too approximate a representation of our percepts and mental models, and that much of our knowledge comes from non-linguistic mental models used in tasks like building or planning actions.

The complexity of the world is difficult to represent in AI models. LeCun references the Moravec Paradox, which highlights the disparity between tasks humans find easy (like recognizing faces) and those that are challenging for AI (like reasoning about the physical world). He argues that LLMs, while able to pass exams, lack the ability to learn tasks that humans and animals can easily master.

LeCun questions whether LLMs can construct a world model that understands intuitive physics and common sense reasoning about physical reality. He believes the current LLMs are not capable of this and explains that LLMs predict the next word in a sequence without planning the entire sentence or understanding the underlying concepts.

LLMs operate by predicting the next word based on previous words, using a probability distribution over possible words. LeCun contrasts this with human thought, which often occurs before language and is not tied to a specific language. He suggests that future AI systems should plan their responses before producing them, unlike LLMs that generate text one word at a time.

LeCun criticizes the generative approach of training AI on video to predict future frames, stating that we cannot represent distributions over high-dimensional continuous spaces effectively. He argues that the world is too complex for AI to predict all details accurately, which is why generative models have failed to learn good representations of images and videos.

Instead of generative models, LeCun advocates for joint embedding architectures (JEPAs), which learn abstract representations without attempting to reconstruct every detail. JEPAs focus on predictable information and discard unpredictable details, lifting the level of abstraction and allowing AI to learn more conceptual representations.

LeCun explains that self-supervised learning works better with redundant data, which is more prevalent in sensory input than in text. He believes that JEPAs will be essential for developing intelligent systems that can understand the world and plan actions based on abstract representations.

The future of AI, according to LeCun, involves JEPAs that can learn from video and other sensory inputs to develop world models. These models will enable AI to predict the consequences of actions and plan accordingly, moving towards human-level intelligence.

LeCun is skeptical of autoregressive LLMs, believing they cannot achieve deep understanding or reasoning. He predicts that future AI systems will use objective-driven architectures to plan responses before converting them into language, a process that will be more efficient and capable of complex reasoning.

LeCun envisions a future where AI systems are diverse, open-source, and capable of

embodying various cultural and linguistic backgrounds. He emphasizes the importance of open-source AI for democracy, cultural preservation, and the development of an AI industry that serves a wide range of needs and values.

LeCun is excited about the potential for AI to make humanity smarter, comparing the impact of AI to the invention of the printing press. He believes AI will amplify human intelligence and enable everyone to have access to smart AI assistants, leading to a more informed and intelligent society.

LeCun dismisses the fears of AI doomers, arguing that the emergence of superintelligence will not be a sudden event but a gradual process. He believes that AI systems will be designed to be controllable and safe, with guardrails in place to prevent rogue AI from causing harm.

LeCun is optimistic about the future of robotics, predicting that the next decade will see significant advancements in robots that can understand the world and perform complex tasks. He stresses the need for AI systems that can learn world models and plan hierarchically to enable the development of versatile and capable robots.

LeCun expresses hope for the future of humanity, believing that AI will enhance human intelligence and contribute to the betterment of society. He trusts that people are fundamentally good and that AI, particularly open-source AI, will empower the positive aspects of human nature.